



Tutorial 2

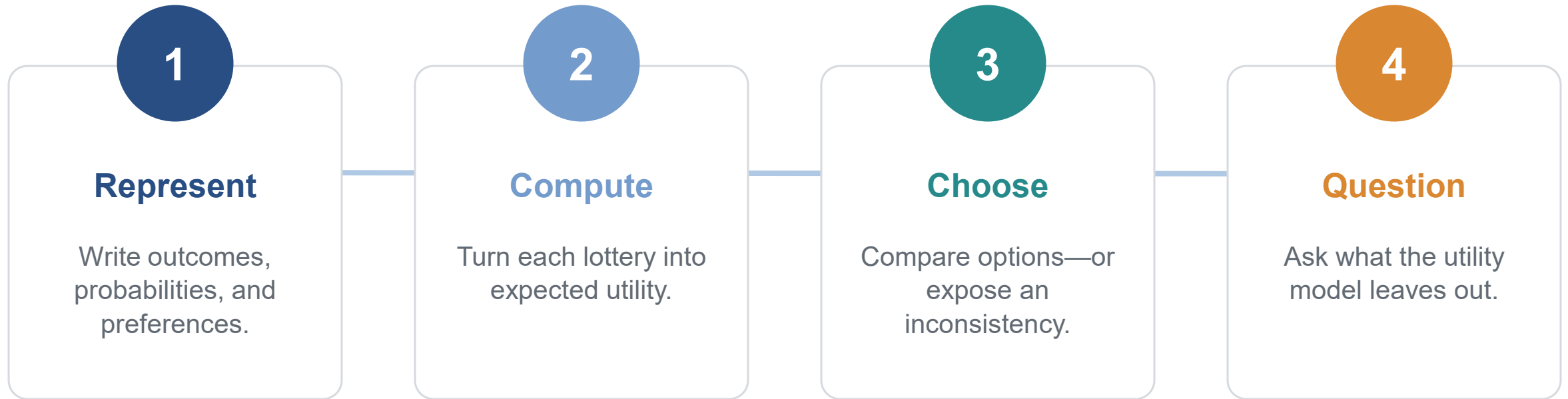
Rational Decision Making

Preferences • Expected utility • Risk • Allais paradox

jing.yu@nus.edu.sg

choose $a^* = \arg \max_a \sum_s P(s | a) U(s)$

■ One chain connects all three problems





01 Review

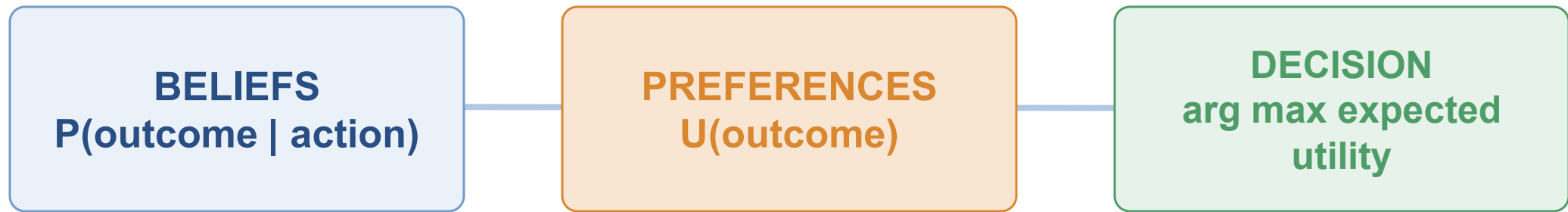
The minimum theory needed to solve every question.

QUESTION → DISCUSSION → REVEAL

■ Rational choice needs beliefs and preferences

THINK • PAIR • SHARE

If an agent knows every probability but has no preferences, can it choose rationally?



Probability says what may happen. Utility says how much the outcome matters.

- MEU combines the two inputs into one comparison

ANSWER / DISCUSS

$$EU(a) = \sum_s P(\text{Result}(a)=s) U(s)$$

$$a^* = \arg \max_a EU(a)$$

Belief error changes the weights. Utility error changes the values. Both can change the chosen action.

■ A lottery packages uncertain outcomes

Lecture pp. 16–17

Notation

$$L = [p_1, s_1; p_2, s_2; \dots; p_n, s_n]$$

- $p_i \geq 0$ and $\sum_i p_i = 1$
- s_i is a final outcome
- $[1, x]$ is a degenerate lottery: outcome x for sure

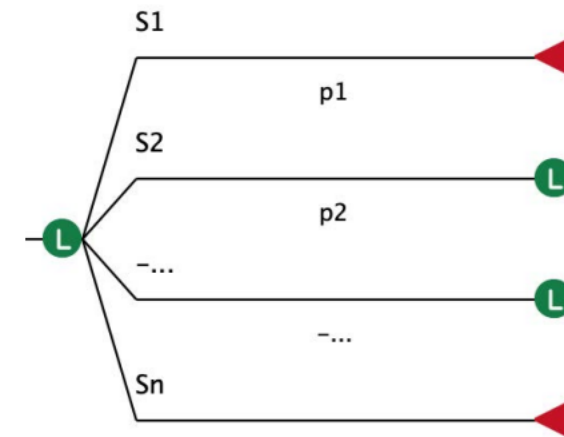


Diagram adapted from the current lecture slides, pp. 16–17.

■ Quick check: read the notation before calculating

CALL OUT THE ANSWER

$$L = [0.3, 8; 0.7, 2]$$

1

What are the outcomes?

2

Which outcome is more likely?

3

Is this a degenerate lottery?

Check: probabilities sum to 1 before doing anything else.

■ Expected utility weights utility—not money

$$EU(L) = \sum_i p_i U(s_i)$$

Same lottery, three attitudes

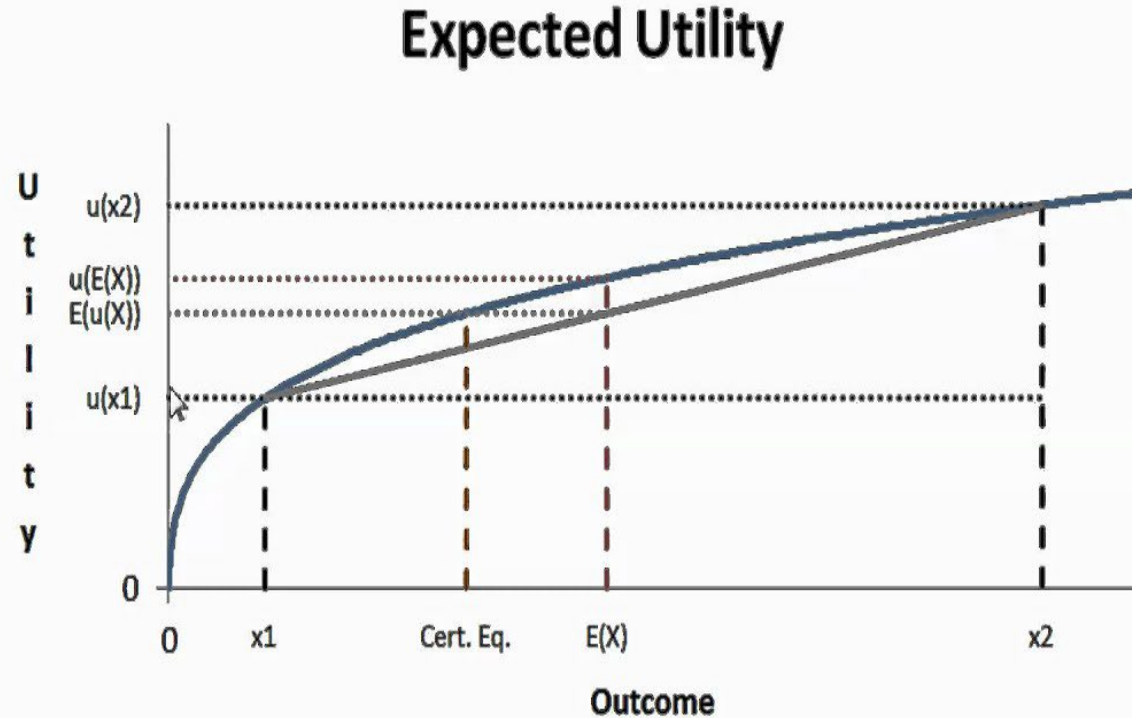
Concave U
certainty is relatively
attractive
(risk-averse)

Linear U
EU = expected money
(risk-neutral)

Convex U
spread is relatively
attractive
(risk-seeking)

■ Curvature turns wealth into risk attitude

AIMA 4e Fig. 16.2



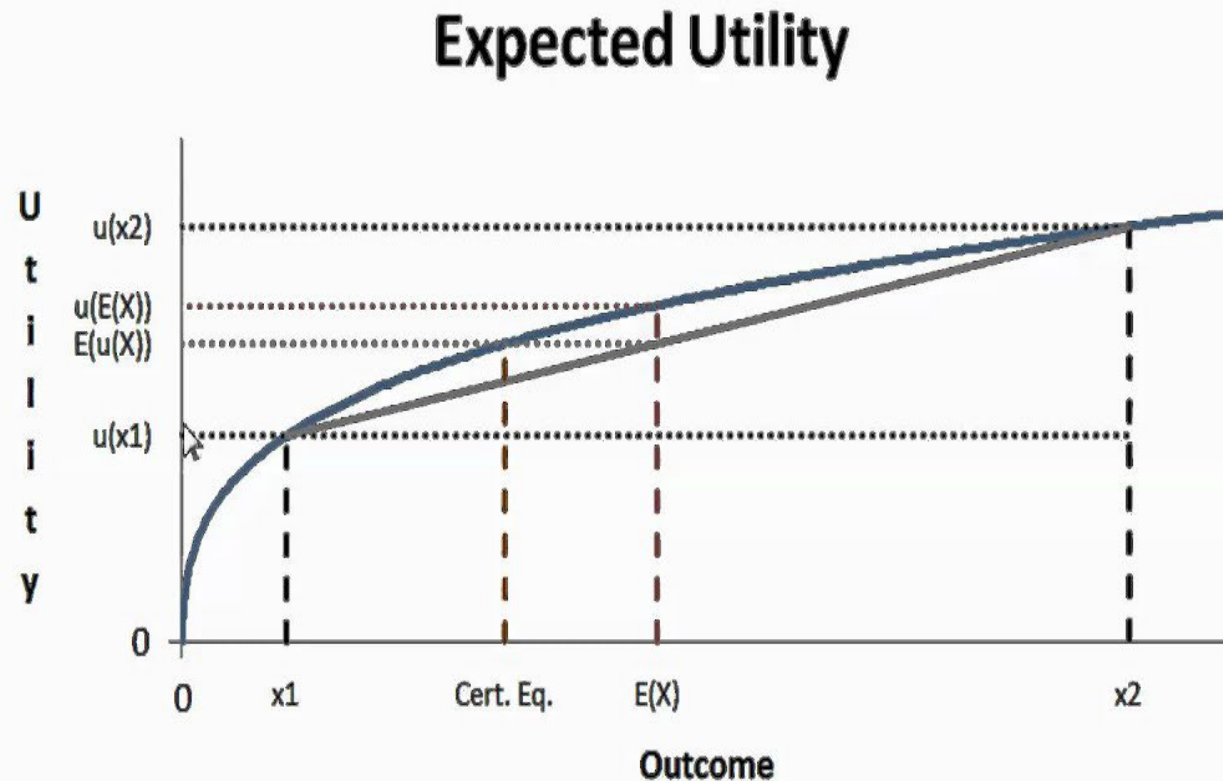
Positive wealth

- Concavity means diminishing marginal utility
- A bad loss hurts more than an equal gain helps
- A sure outcome can beat a higher-EMV lottery

$$\text{concave } U \Rightarrow U(E[X]) \geq E[U(X)]$$

AIMA 4e, Figure 16.2 and §16.3.2 (utility of money).

■ What is CE?



$$U(\text{CE}) = \mathbb{E}[U(X)]$$

$$\text{CE} = U^{-1}(\mathbb{E}[U(X)]) .$$

The Certainty Equivalent is the guaranteed amount that feels just as valuable as taking the gamble.

Risk Premium?

The risk premium is the amount a person is willing to give up to replace a risky gamble with certainty.

$$\text{Risk premium} = \mathbb{E}[X] - \text{CE}$$

$$U(\text{CE}) = \mathbb{E}[U(X)]$$

so:

$$\text{CE} = U^{-1}(\mathbb{E}[U(X)]) .$$

In the diagram:

1. Calculate the lottery's expected utility, $\mathbb{E}[U(X)]$.
2. Move horizontally to the utility curve.
3. Move downward to the outcome axis.
4. That outcome is the certainty equivalent.

Intuitively, CE answers:

“What guaranteed amount would make me indifferent between taking it and playing the lottery?”

For example, suppose X gives \$0 or \$100 with equal probability and $U(x) = \sqrt{x}$:

$$\mathbb{E}[U(X)] = 0.5\sqrt{0} + 0.5\sqrt{100} = 5.$$

Therefore:

$$\sqrt{\text{CE}} = 5 \quad \Rightarrow \quad \text{CE} = \$25.$$

$$\boxed{\text{Risk premium} = \mathbb{E}[X] - \text{CE}} = 50 - 25 = 25$$



02 Problem 1

Utility Matchmaker: infer the curve from preferences.

QUESTION → DISCUSSION → REVEAL

■ Problem 1 — full question (Part A, 1/2)

OFFICIAL QUESTION •
VERBATIM

Utility Matchmaker: The Case of the Mysterious Preferences

Consider the following lotteries, where $[p, x]$ denotes outcome x with probability p :

$$L_1 = [1, 1], \quad L_2 = [0.5, 2; 0.5, 0], \quad L_3 = [1, 2].$$

Four students have expressed their preferences over these lotteries:

- Adam is indifferent between L_2 and L_1 .
- Bing prefers L_1 to L_2 .
- Candy is indifferent between L_3 and L_2 .
- Diana prefers L_2 to L_1 .

Question text: current AY2627 Tutorial 2, p. 1.

■ Problem 1 — full question (Part A, 2/2)

OFFICIAL QUESTION • VERBATIM

Match each student with a utility function that is consistent with their stated preferences (each student has a different one):

$$L_1=[1,1] \quad \bullet \quad L_2=[0.5,2; 0.5,0] \quad \bullet \quad L_3=[1,2]$$

Adam: $L_2 \sim L_1$ • Bing: $L_1 > L_2$
Candy: $L_3 \sim L_2$ • Diana: $L_2 > L_1$

$$(a) \quad U(x)=x^2$$

$$(b) \quad U(x)=x$$

$$(c) \quad U(x)=\sqrt{x}$$

(d) None of the above

Everything needed is here: calculate each EU, then match the orderings.

■ Part A: build one table before matching names

COMPUTE FIRST

$L_1=[1,1]$ • $L_2=[0.5,2; 0.5,0]$ • $L_3=[1,2]$

Adam: $L_2 \sim L_1$ • Bing: $L_1 > L_2$
Candy: $L_3 \sim L_2$ • Diana: $L_2 > L_1$

Utility	EU(L_1)	EU(L_2)	EU(L_3)
$U(x)=x^2$	?	?	?
$U(x)=x$	?	?	?
$U(x)=\sqrt{x}$	?	?	?

Fill the nine EU cells, then match each row's ordering to the four students.

$$EU(a) = \sum_s P(\text{Result}(a)=s) U(s)$$

- Candy forces an equality none of the curves satisfy

REVEAL 4 / 4

$$\begin{aligned} L_3 &= [1, 2] \Rightarrow EU(L_3) = U(2) \\ L_2 &= [0.5, 2; 0.5, 0] \Rightarrow EU(L_2) = 0.5U(2) + 0.5U(0) \end{aligned}$$

Candy requires $L_3 \sim L_2$:

$$\Rightarrow U(0) = U(2)$$

None of x^2 , x , or $\sqrt{x}$ has equal utility at 0 and 2.

■ Problem 1 — full question (Part B)

OFFICIAL QUESTION •
VERBATIM

Assume Agent 1 has utility U_1 and Agent 2 has utility U_2 . If

$$U_1 = k_1 U_2 + k_2, \quad k_1 > 0, \quad k_2 \in \mathbb{R},$$

For $L=[p_i, x_i]$, use $\sum_i p_i=1$ and $EU_j(L)=\sum_i p_i U_j(x_i)$.

do Agents 1 and 2 have the same preferences?

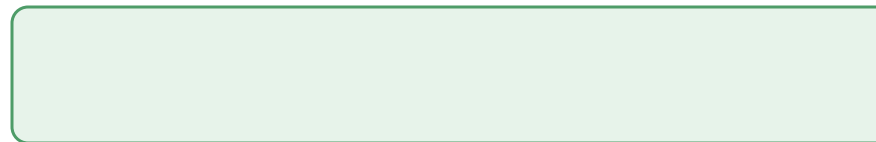
Compare two arbitrary lotteries L and L' ; explain where $k_1 > 0$ is used.

■ Problem 1 Part B: prove invariance for any lotteries

ANSWER / DISCUSS

$$U_1 = k_1 U_2 + k_2, \quad k_1 > 0$$

$$EU_1(L) > EU_1(L')$$



Yes—Agents 1 and 2 have the same preferences.

■ Diagnostic extension: why insist on $k_1 > 0$?

THINK • PAIR • SHARE

Given $U_1 = k_1 U_2 + k_2$, compare lottery rankings when $k_1 < 0$ and when $k_1 = 0$.

$k_1 < 0$

$k_1 = 0$

Only a positive scale factor preserves direction and distinctions.



03 Problem 2

PacBaby: expected utility can justify insurance with lower expected wealth.

[QUESTION](#) → [DISCUSSION](#) → [REVEAL](#)

■ Problem 2 — full question (scenario + part a)

OFFICIAL QUESTION •
VERBATIM

Adventures of PacBaby

PacBaby just found a \$100 note — it is the only thing she owns. Monsters are kind enough not to kill PacBaby, but whenever they find her, they steal all her money. The probability that the monsters find PacBaby is 20% (i.e., 0.2).

PacBaby's utility function is

$$U(x) = \log(1+x)$$

where $\log$ denotes the natural logarithm and x is her monetary wealth. Thus, $U(100) = \log(101)$ and $U(0) = \log(1+0) = 0$.

(a) What is PacBaby's expected utility without insurance?

Question text: current AY2627 Tutorial 2, p. 3.

■ Part (a): without insurance, weight both final utilities

REVEAL a

No theft • probability 0.8
wealth = \$100 • utility

Theft • probability 0.2
wealth = \$0 • utility

■ Problem 2 — full question (parts b–c)

OFFICIAL QUESTION •
VERBATIM

Given: initial wealth \$100 • $P(\text{theft})=0.2$ • $P(\text{no theft})=0.8$ • $U(x)=\log(1+x)$

(b) PacIncome offers theft insurance: for a premium of \$30, PacBaby will be reimbursed \$70 if monsters steal her money. After paying the premium, if no theft occurs, PacBaby holds \$70; if theft occurs, she also ends up with \$70 (the reimbursement). What is PacBaby's expected utility if she buys insurance? Should she buy it to maximize expected utility?

Question text: current AY2627 Tutorial 2, p. 3.

■ Part (b): insurance makes utility state-independent

REVEAL b • CALCULATION

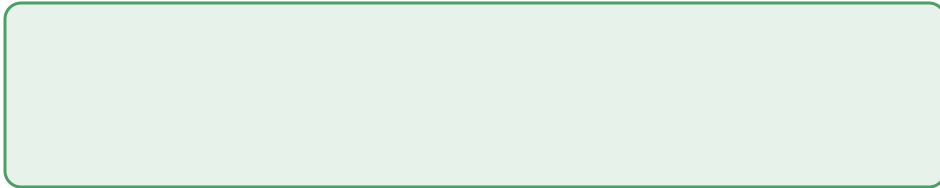
No theft • probability 0.8
wealth = \$70

Theft • probability 0.2
wealth = \$70

■ PacBaby buys insurance—even though expected wealth falls

REVEAL b • DECISION

Expected utility



Expected wealth with v.s. without insurance?

$\$70 < \80

**Choose by expected utility, not expected dollars.
The uninsured \$0 outcome is very painful under log utility.**

Decision:??

■ Problem 2 — full question (parts b–c)

OFFICIAL QUESTION •
VERBATIM

Given: initial wealth \$100 • $P(\text{theft})=0.2$ • $P(\text{no theft})=0.8$ • $U(x)=\log(1+x)$

(b) PacIncome offers theft insurance: for a premium of \$30, PacBaby will be reimbursed \$70 if monsters steal her money. After paying the premium, if no theft occurs, PacBaby holds \$70; if theft occurs, she also ends up with \$70 (the reimbursement). What is PacBaby's expected utility if she buys insurance? Should she buy it to maximize expected utility?

(c) From PacIncome's point of view, what is the expected monetary value of selling this insurance?

Track final wealth for PacBaby and cash flow for PacIncome separately.

Question text: current AY2627 Tutorial 2, p. 3.

■ Part (c): the insurer has positive expected profit

REVEAL c

No theft • probability 0.8
PacIncome keeps premium: +\$30

Theft • probability 0.2
Collect \$30, pay \$70: -\$40

EMV =

■ Extension: PacBaby's certainty equivalent is only about \$39.13

Without insurance

$$EU = 0.8 \log(101) = 3.6921$$

$$\log(1+CE)=3.6921$$

$$CE = e^{3.6921} - 1 \approx \$39.13$$

Interpretation

The uninsured lottery has EMV \$80 but feels equivalent to only \$39.13 for PacBaby.

$$\text{risk premium} \approx 80 - 39.13 = \$40.87$$

Extension: What is the max amount PacBaby is willing to pay for insurance?

A general insurance model is:

$$\text{Final wealth} = W - P - L + I(L),$$

where:

- W : starting wealth
- P : premium
- L : loss from theft
- $I(L)$: insurance payout

With full insurance, $I(L) = L$, so:

$$\text{Final wealth} = W - P$$

in either state.

1. We assume full insurance payout. So with 0 uncertainty, the final wealth is $W - P$

If insurance costs P , PacBaby receives guaranteed final wealth:

$$100 - P.$$

At the maximum acceptable premium, PacBaby must be indifferent between:

- Remaining uninsured, worth \$39.13 for sure to her.
- Buying insurance and receiving $100 - P$ for sure.

Therefore, at indifference:

$$100 - P = 39.13.$$

Hence:

$$P_{\max} = 60.87.$$

2. The certainty equivalent is the guaranteed dollar amount that PacBaby values as much as the uninsured gamble. Here, the gamble has a certainty equivalent of \$39.13. Therefore, the maximum premium leaves her with exactly \$39.13 in guaranteed final wealth:
 $100 - P_{\max} = 39.13$

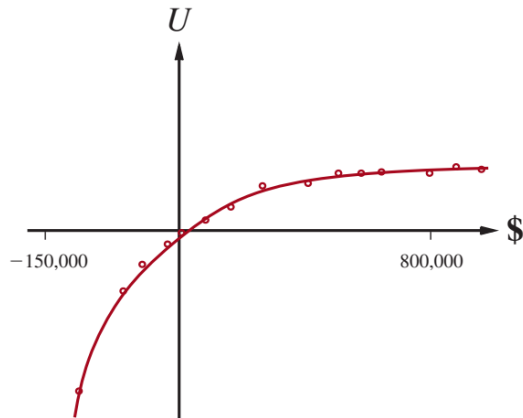
■ Why can both sides benefit? Money out of nowhere?

EXPLAIN IN WORDS

Facts: PacBaby has $U(x)=\log(1+x)$; uninsured wealth is \$100 (0.8) or \$0 (0.2); insured wealth is \$70 for sure. PacIncome earns +\$30 or -\$40.

PacBaby

- Concave utility
- A catastrophic \$0 outcome dominates
- Pays to smooth wealth to \$70



PacIncome

- Diversifies across many policies
- Policy has positive EMV
- Pools idiosyncratic theft risk

Risk transfer can create value even when no money appears from nowhere.



04 Problem 3

Allais paradox: when observed choices cannot share one expected-utility representation.

QUESTION → DISCUSSION → REVEAL

■ Problem 3 — full question (setup)

OFFICIAL QUESTION • VERBATIM

The Allais paradox (Allais, 1953) is a well-known problem potentially suggesting that humans are “predictably irrational” (Ariely, 2009). People are given a choice between lotteries A and B and then between C and D, which have the following prizes:

A: 80% chance of \$4000; otherwise \$0

C: 20% chance of \$4000; otherwise \$0

B: 100% chance of \$3000

D: 25% chance of \$3000; otherwise \$0

Most people consistently prefer B over A (i.e., taking the sure payoff), and C over D (taking the higher EMV).

Question text: current AY2627 Tutorial 2, p. 4.

■ Problem 3 — full question (part a)

OFFICIAL QUESTION •
VERBATIM

A: $.8 \rightarrow \$4000$; $.2 \rightarrow \$0$

B: \$3000 for sure

C: $.2 \rightarrow \$4000$; $.8 \rightarrow \$0$

D: $.25 \rightarrow \$3000$; $.75 \rightarrow \$0$

Observed judgments: $B \succ A$ and $C \succ D$. Normalize $U(\$0)=0$.

(a) Show that the normative analysis (i.e., describing how a rational agent should act) disagrees.

Hint: Set $U(\$0)=0$; show that the preferences between A, B and C, D are opposites, hence a contradiction. Use an arbitrary Utility Function

Translate each observed preference into an expected-utility inequality before manipulating it.

■ Vote twice—then defend the pair

ANONYMOUS SHOW OF HANDS

Choice 1

A
80% \$4000
20% \$0

B
\$3000 for sure

Choice 2

C
20% \$4000
80% \$0

D
25% \$3000
75% \$0

Can one expected-utility function rationalize your two choices at once?

Common empirical pattern: $B \succ A$ and $C \succ D$.

■ Part (a), first choice: $B \succ A$

REVEAL a • STEP 1

$$EU(A) = 0.8U(\$4000) + 0.2U(\$0)$$

$$EU(B) = U(\$3000)$$

Set $U(\$0) = 0$. Then $B \succ A$ implies

$$U(\$3000) > 0.8U(\$4000)$$

■ Part (a), second choice: $C \succ D$

REVEAL a • STEP 2

$$EU(C) = 0.2U(\$4000) + 0.8U(\$0)$$

$$EU(D) = 0.25U(\$3000) + 0.75U(\$0)$$

With $U(\$0)=0$, $C \succ D$ implies

$$0.2U(\$4000) > 0.25U(\$3000)$$

■ The two observed preferences demand opposite inequalities

REVEAL a • CONTRADICTION

$$B \succ A \Rightarrow$$

$$C \succ D \Rightarrow$$

■ Problem 3 — full question (part b)

OFFICIAL QUESTION •
VERBATIM

A: $.8 \rightarrow \$4000$; $.2 \rightarrow \$0$

B: \$3000 for sure

C: $.2 \rightarrow \$4000$; $.8 \rightarrow \$0$

D: $.25 \rightarrow \$3000$; $.75 \rightarrow \$0$

Observed judgments: $B \succ A$ and $C \succ D$.

(b) Prove that the judgments $B \succ A$ and $C \succ D$ in the above Allais paradox violate the axiom of substitutability.

Hint: You may wish to consider using the axiom of decomposability.

Two axioms do the heavy lifting in Allais

AIMA 4e Fig. 16.1

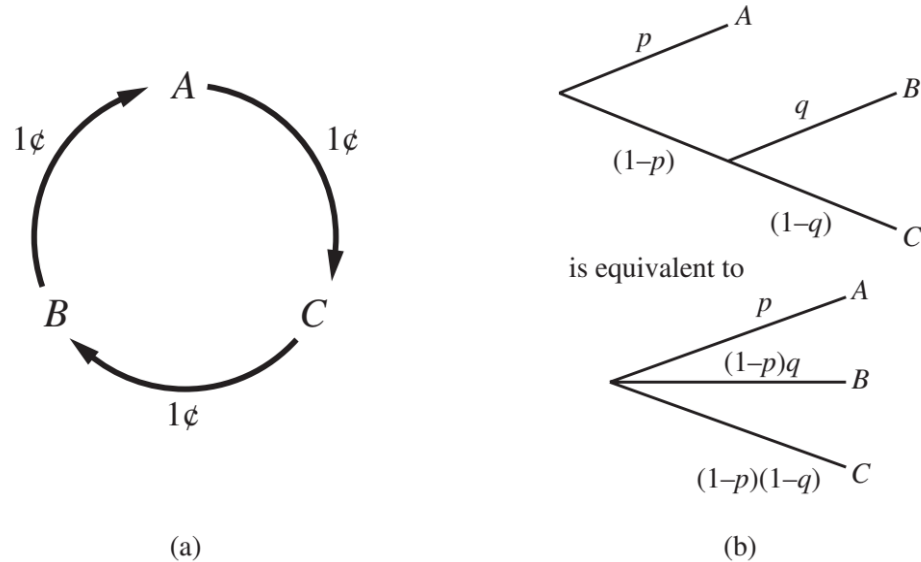


Figure 16.1 (a) Nontransitive preferences $A \succ B \succ C \succ A$ can result in irrational behavior: a cycle of exchanges each costing one cent. (b) The decomposability axiom.

- **Decomposability:** Compound lotteries can be reduced to simpler ones using the laws of probability. This has been called the “no fun in gambling” rule: as Figure 16.1(b)

Decomposability

Decomposability

Flatten a compound lottery: only final probabilities matter.

Substitutability

If $A \succ B$, replacing A by B inside the same lottery must make it worse.

$$A \succ B \Rightarrow [p, A; 1-p, C] \succ [p, B; 1-p, C]$$

■ Part (b): decomposability reveals the shared structure

Relate A and C, Relate B and D

REVEAL b • STEP 1

A itself is an 80–20 lottery. Put A inside a 25% outer branch:

$$[0.25, A; 0.75, \$0] \sim C$$

because $0.25 \times 0.80 = 0.20$ chance of \$4000.

Observed judgments: $B \succ A$ and $C \succ D$.

A: .8→\$4000; .2→\$0

B: \$3000 for sure

C: .2→\$4000; .8→\$0

D: .25→\$3000; .75→\$0

■ Substitutability predicts the opposite ordering

REVEAL b • STEP 2

$C \succ D$

and decomposability imply

Observed judgments: $B \succ A$ and $C \succ D$.

A: $.8 \rightarrow \$4000$; $.2 \rightarrow \$0$

B: \$3000 for sure

C: $.2 \rightarrow \$4000$; $.8 \rightarrow \$0$

D: $.25 \rightarrow \$3000$; $.75 \rightarrow \$0$

■ Diagnosis: the common Allais pattern violates substitutability

FINAL ANSWER

Observed

$B \succ A$ and $C \succ D$

Expected-utility axioms imply

$B \succ A \Rightarrow D \succ C$

The shared consequence (\$0 with probability 0.75) should not reverse the preference between A and B. Yet it does.

That reversal is a violation of...?

■ Allais Paradox: which axiom does the contradiction expose?

VOTE FIRST • WHICH AXIOM?

A1 Orderability Any pair can be compared.

A2 Transitivity $A \succ B$ and $B \succ C \Rightarrow A \succ C$.

A3 Continuity $A \succ B \succ C \Rightarrow B \sim [p, A; 1-p, C]$.

A4 Independence Substitution in the same mixture preserves a ranking.

A5 Monotonicity More chance of the better prize is better.

A6 Decomposability Flatten a compound lottery.

PART (a) • INCOMPATIBLE EU INEQUALITIES

A: $.8 \rightarrow \$4k, .2 \rightarrow \0 B: $\$3k$ for sure
C: $.2 \rightarrow \$4k, .8 \rightarrow \0 D: $.25 \rightarrow \$3k, .75 \rightarrow \0

Observed: $B \succ A$ and $C \succ D$

$B \succ A \Rightarrow U(\$3k) > 0.8U(\$4k)$

$C \succ D \Rightarrow 0.8U(\$4k) > U(\$3k)$

No single expected-utility function satisfies both.

PART (b) • NORMATIVE ANALYSIS

A6 lets us rewrite the second pair:

$C \sim [0.25, A; 0.75, \$0]$

$D \sim [0.25, B; 0.75, \$0]$

$B \succ A \Rightarrow D \succ C$ but observed $C \succ D$

A4 says $B \succ A$ must remain B-better after both are mixed with the same 75% chance of \$0:

■ Why does this happen?

One explanation for the apparently irrational preferences is the **certainty effect** (Kahneman and Tversky, 1979): people are strongly attracted to gains that are certain. There are several reasons why this may be so.

First, people may prefer to reduce their computational burden; by choosing certain outcomes, they don't have to compute with probabilities. But the effect persists even when the computations involved are very easy ones.

Second, people may distrust the legitimacy of the stated probabilities. I trust that a coin flip is roughly 50/50 if I have control over the coin and the flip, but I may distrust the result if the flip is done by someone with a vested interest in the outcome.⁵ In the presence of distrust, it might be better to go for the sure thing.⁶

Third, people may be accounting for their emotional state as well as their financial state. People know they would experience regret if they gave up a certain reward (B) for an 80% chance at a higher reward and then lost.

In other words, if A is chosen, there is a 20% chance of getting no money *and feeling like a complete idiot*, which is worse than just getting no money. So perhaps people who choose B over A and C over D are not irrational; they are willing to give up \$200 of EMV to avoid a 20% chance of feeling like an idiot.

AIMA's interpretation

- Normative: how an ideal rational agent should act
- Descriptive: how people actually choose
- Certainty, distrust, regret, and framing may enter the real outcome
- A richer state/utility model may explain part of the gap

Normative inconsistency invites scientific questions

AIMA 4e, §16.3.4, printed p. 538 (Allais paradox and certainty effect).



05 Research directions

Utility is increasingly treated as learned, uncertain, interactive, and stress-tested.

QUESTION → DISCUSSION → REVEAL

■ Three research questions students can explain without RL

1

Learn the utility curve

Which $U(x)$ best explains a person's choices?

2

Test new choices

Does the learned ranking survive unfamiliar options?

3

LLMs' Risk Profile

Do LLM choices obey the same decision principles?

All three reuse the tutorial pipeline: lottery → expected utility → preference.

■ Project 1: learn a person's utility curve from choices in Inverse RL

BACKGROUND

Inverse RL is the topic of learning a Reward Function from data. So far, people assume Risk Neutral Agent.
NO inference of risk attitudes

People can rank the same lotteries differently because the shape of $U(x)$ —not only the money—captures their attitude to risk.

PROBLEM

Given a small set of observed choices, can we identify whether the person is risk-neutral, risk-averse, or risk-seeking?

PROPOSED CONTRIBUTION

Fit simple candidate curves such as x , $\sqrt{x}$, and $\log(1+x)$, then test which curve predicts choices on new lotteries.

$$EU_{\theta}(L) = \sum_i p_i U_{\theta}(x_i)$$

Paper seed: Lazzati & Metelli, ICML 2025
<https://proceedings.mlr.press/v267/lazzati25a.html>

Direct tutorial links: matching utility functions; risk aversion; expected utility.

■ Project 2: test whether learned preferences work on new options in Reward Models learned from Human Feedback

BACKGROUND

A preference model learns a ranking from labelled pairs: option A is preferred to option B.

PROBLEM

A model may rank familiar examples well but fail when the system starts producing options unlike those in its training data.

PROPOSED CONTRIBUTION

Create familiar and unfamiliar lottery sets. Compare random questions with questions chosen near the system's current choices.

Key test: high accuracy on familiar choices can coexist with poor ranking of new choices.

Paper seed: Zhu et al., NeurIPS 2025

https://proceedings.neurips.cc/paper_files/paper/2025/hash/9cd2a03f427acc03b6ddb9c8f3be57c-Abstract-Conference.html

Tutorial link: a useful utility representation must preserve the correct ordering of lotteries.

■ Project 3: Can LLM Capture Human Preference? What is LLMs own Risk Preference?

BACKGROUND

LLMs increasingly give advice when outcomes are uncertain, but their displayed choices can depend on wording and context.

PROBLEM

Do their answers preserve rankings under harmless transformations? Do gain/loss frames or Allais-style pairs cause reversals?

PROPOSED CONTRIBUTION

Build matched lottery prompts, repeat each prompt, and report choice frequencies, axiom violations, and framing effects.

$$U'(x) = aU(x) + b, a > 0 \Rightarrow \text{same ranking}$$

Measure behavior under a protocol—not a stable ‘personality.’

*Preprint seeds: Song et al. (2025); <https://arxiv.org/pdf/2506.23107>
Payne (2025). <https://arxiv.org/abs/2508.00902>*

**LLMS are trained on Human Data and Human Feedback.
Does it inherit Human Risk Attitudes?**

■ The papers behind the three ideas—in one minute

Lazzati & Metelli
ICML 2025

Learn risk-sensitive utility from demonstrations. One setting may not reveal a unique utility; observing choices across settings can help.

PEER
REVIEWED

Zhu et al.
NeurIPS 2025

Static preference data may not cover the new options created by a changing policy. Querying nearby unfamiliar options improves coverage.

PEER
REVIEWED

Song et al. + Payne
2025

LLM risk choices vary across model, language, framing, and scenario. Useful motivation for an audit, but not settled evidence about cognition.

PREPRINTS

For a class project: simplify the setting, keep the research question, and make the failure test explicit.